

The Research Content

of Investment Strategies

May 12, 2026

Red text marks placeholders or tasks before circulation. Purple text marks temporal quantities, current sample facts, or claims likely to change as the measurement design updates.

Abstract

Academic finance and investable products often describe related economic ideas in different languages. Academic papers study mechanisms, strategies, risks, and empirical relationships. Product documents describe objectives, exposures, instruments, benchmarks, rules, and investor-facing risks. This note develops a measurement design for comparing the two. The product side is represented as a small design graph, and the academic side is represented through nearby research-graph neighborhoods.

[**Placeholder** – replace this abstract placeholder with the main empirical finding once the reviewed product–research link sample is fixed. The finding should state which product families have close strategy links, which connect through mechanisms or market structures, and which remain boundary cases, using source-checked evidence rather than the current first-pass labels.]

1 Introduction

Investment products often look familiar to academic finance, but only after translation. A covered-call fund looks close to the buy-write strategies studied in the literature. A target-date fund looks close to lifecycle portfolio choice. A buffer ETF is less direct: the product describes reference assets, FLEX options, caps, buffers, and outcome periods, while the academic neighborhood is closer to portfolio insurance and option-based payoff protection. These examples share finance ideas, but the product and the paper rarely describe the idea in the same form.

This paper studies that translation. Academic finance describes mechanisms, strategies, risks, and empirical relationships. Product documents describe objectives, exposures, instruments, benchmarks, payoff rules, horizons, and investor-facing risks. The research question is whether those two descriptions can be compared in a disciplined way. When an investment strategy is sold as a product, how much of its design can be linked to academic finance, and at what level: the same named strategy, the same mechanism, a broader design pattern, a broad concept, or no credible link?

[Placeholder – add the final introduction result once the reviewed sample is fixed. This paragraph should report the main empirical pattern in product–research links, with enough detail to show what is close, what is indirect, and what remains noisy.]

The paper’s measurement object is a product–research link. On the product side, I represent a prospectus as a small design graph. This is simply a compact map of what the product says it does: what it seeks, what it holds or references, which instruments it uses, which rule or horizon defines the payoff, which benchmark it tracks, and which risks it discloses. On the academic side, I use nearby papers and concepts from FrontierGraph, where academic papers are represented through concepts and relationships extracted from text. A product–research link connects the product description to the nearby academic research and classifies the kind of connection.

[Placeholder – insert the final headline examples after source-text review. This should give 3–4 representative product families and state the level at which each connects to academic finance. Avoid presenting the current pilot typology as a final result.]

The question matters for several audiences. For academics, the link asks which research objects travel into market practice and which product objects remain under-

studied. For product designers, it asks whether a strategy is grounded in a studied mechanism or mainly packaged around a theme. For allocators, it separates products that implement a known strategy from products that share only surface vocabulary with the literature. For financial-innovation researchers, it creates a way to ask whether products follow academic formalization, precede it, or evolve in a parallel language.

The contribution is to define and measure a link between academic research objects and product design objects. Existing work on asset pricing, factor investing, funds, and financial innovation gives many reasons to care about how ideas become investable. The missing object is the translation layer itself. A keyword match cannot distinguish a product that discloses volatility risk from one that follows a volatility-control rule. A product graph can make that distinction visible and comparable to academic research graphs.

Table 1: Plain-language objects in the note

Term	Meaning in this note
Product document	The fund filing text that describes what a mutual fund or ETF says it does.
Product graph	A compact description of the product: objective, holdings, instruments, rules, benchmark, horizon, and risks.
Academic neighborhood	The nearby papers, concepts, and relationships in the academic finance graph around the same broad idea.
Product–research link	The connection between a product description and nearby academic finance work.
Link type	The kind of connection: same strategy, same mechanism, broader pattern, concept-only match, or no useful link.

Notes: The table defines project-specific terms before the empirical design uses them. The terms are descriptive measurement objects, not investment ratings or performance measures.

[**Placeholder** – add literature citations and exact positioning. Likely anchors: asset-pricing anomalies and factor investing; mutual funds and ETFs; financial innovation and product design; text-as-data and knowledge graphs in economics and finance; FrontierGraph as the academic graph source.]

[**Placeholder** – replace the provisional first-pass link counts with final reviewed

counts once source-text review is complete.]

[**Placeholder** – add final roadmap paragraph once the empirical design is chosen. The current preferred path is cross-sectional link measurement first, with product-entry timing as a later empirical layer.]

2 The Research Question

The broad question is:

How do ideas in academic finance appear in investable products, and where do products use ideas that academic finance has not yet described in the same form?

There are several possible versions of this question.

1. A product may directly implement an academic strategy.
2. A product may use a mechanism that academic finance studies, even if the product wrapper is newer or more specific.
3. A product may share only a broader pattern with academic work, such as rule-based exposure transformation or state-dependent allocation.
4. A product may mainly share words with the literature, without a substantive link.
5. A product may be industry-led, with little nearby academic treatment at the time it appears.

[**Placeholder** – add 2–3 concrete examples for each link type once the family-level review is stable.]

3 Data Objects

The product side comes from regulatory filings for U.S. mutual funds and exchange-traded funds. These filings include the text that a fund uses to describe itself to investors. The most useful fields for this project are the investment objective, principal investment strategy, benchmark or index description, and principal risks. These fields are useful because they usually say what the fund is trying to deliver, which assets

or instruments it uses, which index or reference asset matters, and what risks the investor is asked to bear.

The unit on the product side is a product document. A document is tied to a fund or fund series and a filing date. It may describe a broad index fund, a target-date fund, a covered-call ETF, a buffer ETF, a crypto-linked product, or another strategy. I treat the document as a source of product design information, not as marketing copy to be scored as good or bad.

The academic side comes from FrontierGraph. FrontierGraph is a graph representation of academic research. Instead of treating a paper only as a title, abstract, citation count, or journal publication, it represents the paper as a local graph. The nodes are research concepts, such as volatility, covered calls, target-date funds, liquidity, portfolio insurance, or bitcoin futures. The edges are relationships among those concepts, such as one object predicting another, one mechanism explaining another, or one variable affecting another. This representation makes academic papers comparable at the level of ideas and relationships.

The comparison uses both objects. A product document is turned into a product design graph. An academic paper, or a nearby set of academic papers, is represented through a research graph neighborhood. The link asks whether the product design and the academic neighborhood describe the same strategy, the same mechanism, a looser broader pattern, only a shared concept, or no credible connection.

Add final sample facts here once the product-family screen, graph-extracted sample, and reviewed link sample are fixed.

[**Placeholder** – insert a table showing the current product sample construction: source filings, candidate screens, extracted links, reviewed links, and usable examples.]

[**Placeholder** – add the exact FrontierGraph citation and describe the current academic source cut: field coverage, paper universe, graph extraction status, and how finance-relevant neighborhoods are selected.]

[**Placeholder** – define the product launch or first-observed date field if the paper studies timing between research and products.]

[**Placeholder** – document how SEC filing fields are mapped into objective, strategy, benchmark, and risk text, including any exclusions or cleaning rules.]

4 Representing Products As Design Graphs

A product graph is a compact description of what a fund says it does. The graph does not try to summarize the whole prospectus. It extracts the parts that describe the product's design: the objective, the strategy, the instruments, the exposures, the rules, the benchmark, and the risks.

The reason to use a graph is that the same word can play different roles. A prospectus may mention options because the product uses options to create a payoff. It may mention options only as a risk. It may mention volatility because the strategy changes when volatility is high, because the product targets low-volatility stocks, or because volatility appears in a generic risk disclosure. A keyword screen treats these cases as similar. A product graph separates them.

The current product graph uses nodes such as:

- product
- objective
- strategy
- instrument
- reference asset
- benchmark or index
- rule or horizon
- factor or style
- risk

The edges say how those objects relate to the product. The current edge types include:

- product seeks objective
- product uses instrument
- product references asset
- product tracks benchmark
- product has rule or horizon

- strategy targets factor or style
- product discloses risk

For example, a covered-call product might have an income objective, an equity or bitcoin reference exposure, and an option-writing strategy. A target-date fund might have a retirement date, underlying funds, and a glide path that changes the equity and fixed-income mix over time. A buffer ETF might have a reference asset, FLEX options, an outcome period, a cap, and a downside-protection rule. These are not just themes. They are product-design objects.

[**Placeholder** – add one worked product example as a small table or figure: product nodes, product edges, and source quotes.]

[**Placeholder** – add the final product-family screen rule once it is fixed. The rule should explain what evidence in the prospectus is sufficient for a product to enter a family.]

[**Placeholder** – add the final extraction scope: which product families are graph-extracted, why they are included, and what evidence is used from each filing.]

5 Link Types

A link is a claim that a product design object and an academic research object describe the same or related economic idea. The link can be close or loose. This distinction matters because product markets translate research ideas in several ways. Sometimes the product is almost the same object that academics study. Sometimes the product is a wrapper around a mechanism that academics study. Sometimes the connection is only a shared word.

A covered-call fund is a relatively close case. A covered-call strategy holds an asset, such as stocks or a stock index, and sells call options on that asset to earn option premium income.¹ Academic papers often study the same strategy under names such as covered calls or buy-write strategies. The product and the paper are therefore close at the strategy level.

A buffer or defined-outcome ETF is different. These products usually use options to create a payoff over a fixed period: the investor gets some upside up to a cap and

¹A call option gives the buyer the right to buy an asset at a specified price. A covered-call strategy sells call options while holding the underlying asset. The strategy usually gives up some upside in exchange for option premium income.

receives some protection against losses up to a buffer or floor.² Academic papers may not study the exact ETF wrapper, but they do study related mechanisms: portfolio insurance, option replication, downside protection, and structured payoffs. The link is therefore less likely to be the same named strategy and more likely to be a mechanism or rationale link.

The current link vocabulary distinguishes these cases:

Table 2: Working link vocabulary

Link type	Meaning
Same strategy	The product and academic literature describe substantially the same operational strategy.
Same mechanism or rationale	The product uses an economic mechanism studied in the literature, but the product wrapper or implementation differs.
Broader pattern	The product and literature share a broader structural pattern, such as rule-based exposure transformation, but not a close strategy match.
Concept only	The product and literature share a broad concept or keyword, but the connection is not yet substantively credible.
No useful link	The candidate match does not hold under review.

The categories are useful because they prevent a false sense of precision. If a product and a paper both mention volatility, the connection may be strong, weak, or accidental. A volatility-control product that adjusts exposure when volatility changes is closer to academic work on target-volatility rules. A product that merely lists volatility as a risk disclosure has a weaker connection. The link label records that difference.

The same logic applies to academic retrieval. A useful academic counterpart should share a strategy, mechanism, rationale, or graph pattern with the product. The link label is therefore part of the measurement, not an afterthought.

[Placeholder – decide whether “same mechanism” and “same rationale” should remain one category or be separated.]

²A buffer ETF is designed to absorb a specified range of losses over an outcome period, often using option positions. A defined-outcome ETF states an intended payoff range, such as capped upside and partial downside protection, over a fixed horizon.

[**Placeholder** – formalize academic counterpart retrieval from FrontierGraph. The method must state how candidate papers, concepts, and local graph neighborhoods are selected.]

[**Placeholder** – create a reproducible link-label review table with product evidence, academic evidence, link type, and reason.]

6 Empirical Designs

The first empirical design is cross-sectional. For each product family, the paper asks what kind of link connects the product design to academic finance. Some families may have direct academic counterparts. Covered-call products, for example, can be compared to academic work on buy-write or covered-call strategies. Other families may connect through mechanisms rather than names. Buffer products can be compared to academic work on portfolio insurance, option replication, and downside protection, even if the exact ETF wrapper is not the object studied in the paper.

This design gives a simple first outcome: the link type. A product family can be close to academic finance at the strategy level, close at the mechanism level, related through a looser broader pattern, connected only by a broad concept, or not credibly connected. The purpose is to compare product families on the type of academic connection they have, rather than to count whether a keyword appears in both places.

The second design is temporal. Once product entry dates are reliable, the same link object can be used to ask whether academic attention comes before, after, or alongside product entry. That design is attractive because it speaks directly to financial innovation: did academic research formalize the idea before the product appeared, or did product markets develop the object first? The timing design needs more data work before it can carry the paper.

[**Placeholder** – timing design parked for later: this requires reliable product first-observed dates, dated academic neighborhoods, and a rule for whether academic attention is measured by first paper, cumulative graph centrality, or local-neighborhood growth.]

The third design is interpretive. Even before timing is available, the product and academic graphs can show how the same broad finance idea is described differently. Products translate economic ideas into wrappers, payoff rules, benchmarks, income objectives, tax risks, and investor-facing exposures. Academic papers usually describe

mechanisms, expected returns, risk channels, and empirical relationships. Comparing the two descriptions helps explain why a product may be research-linked even when it does not share the exact vocabulary of the paper.

The feasible first version is the cross-sectional and interpretive version. It compares product families by the kind of link they have to academic finance and uses product and academic graph language to explain why the link is close or loose. The timing question can then be added once the product-entry data are strong enough.

[**Placeholder** – construct product-family first-observed dates from SEC series identifiers, fund names, tickers, and filing histories if the timing design enters the paper. Check whether first filing date is close enough to product launch date or whether we need an external launch-date source.]

[**Placeholder** – add the final cross-sectional link dataset definition: product family, product-design evidence, academic counterpart evidence, link type, and reason.]

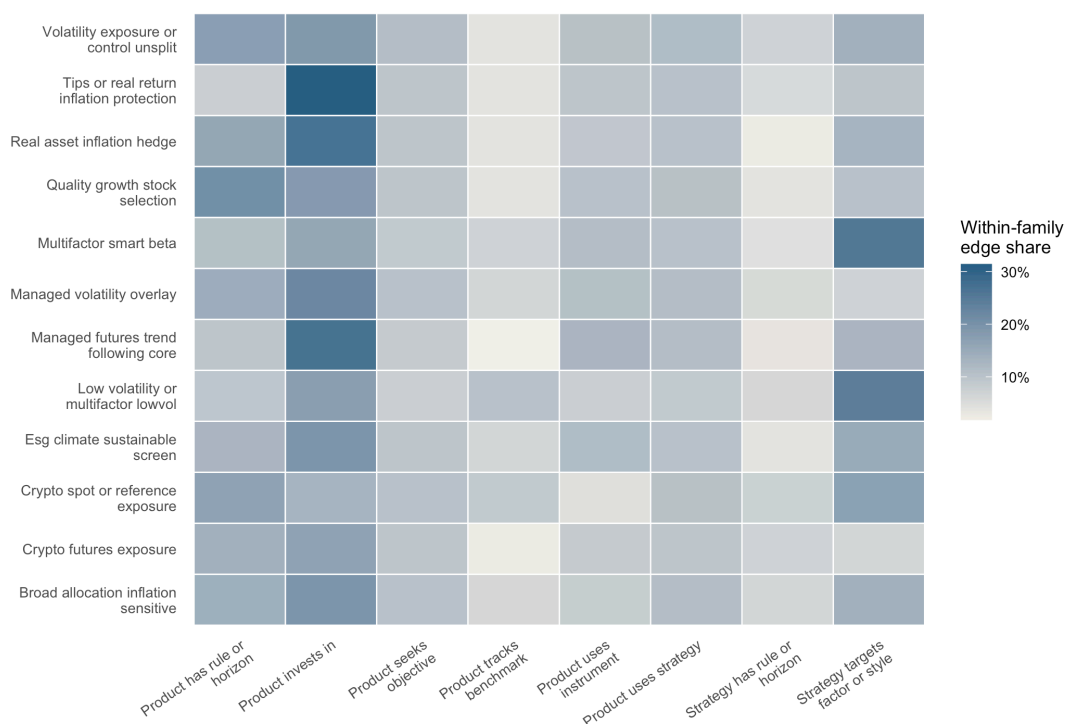
7 Pilot Evidence

The current systematic product-graph tier contains 911 product documents across 12 product families. The extraction produced 9,987 product-side nodes, 9,655 product-side edges, and 7,331 candidate product–research terms, with no parse failures. These quantities are current measurement facts, not final population estimates.

The first lesson is that prospectuses mostly describe product design, not academic mechanisms in miniature. Across the current product families, the recurring objects are holdings and exposures, instruments, objectives, rules and horizons, benchmarks, factor or style terms, and disclosed risks. This is useful. It means the product graph is capturing the language in which products are actually sold: what the fund holds, how it implements the strategy, what rule governs it, and what risk the investor is told to bear.

Figure 1 shows this point at the family level. The heat map reports the share of each family’s extracted edges that belong to the main product-graph relation types. Low-volatility and multifactor products are heavy in factor or style targeting. Crypto, TIPS, real-asset, and broad allocation products are more naturally described through what the product invests in. Managed-volatility and volatility-control products are more rule-like, with many edges connecting the product to a rule, horizon, instrument, or allocation mechanism.

Figure 1: Product-graph signatures differ across product families



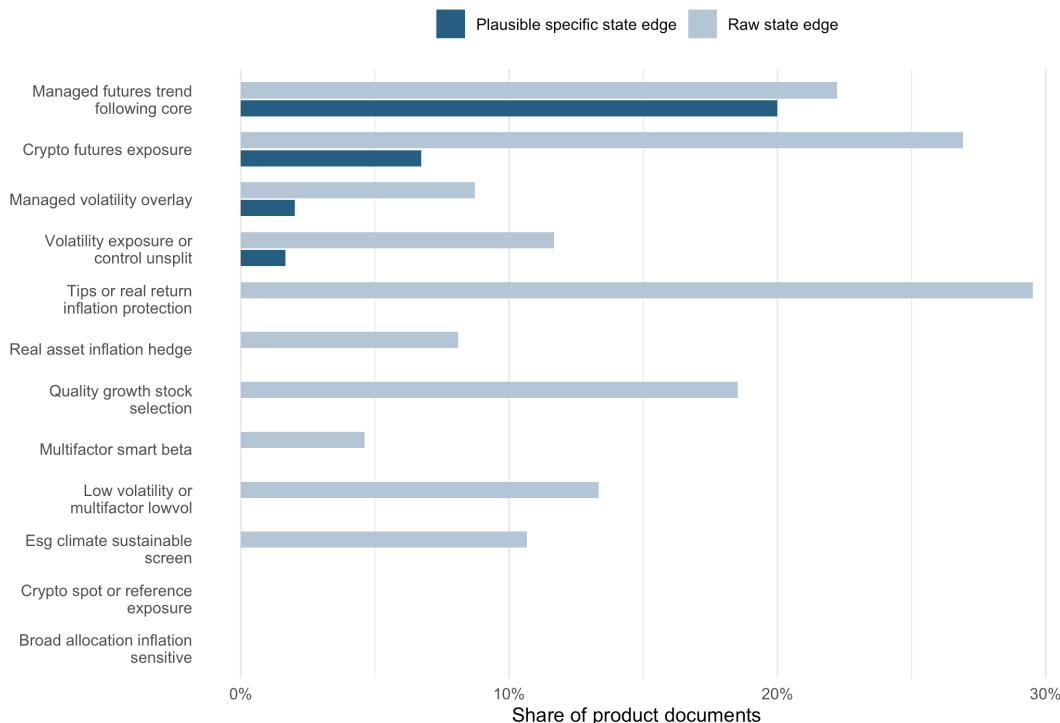
Notes: The figure uses the 911-document systematic product-graph tier. Each cell reports the within-family share of extracted product-graph edges in a given relation type. Darker cells indicate relation types that account for a larger share of the family graph. The figure is descriptive and uses the current extraction schema.

The second lesson is that explicit state-conditioning is present but sparse. The product graph extracted 183 raw state-conditioning edges. A deterministic audit classifies 47 of those as plausible specific state edges, appearing in 29 of the 911 product documents. Most product documents therefore do not say, in a direct way, that the strategy changes under a named market state. They more often describe a design or exposure that may be linked to a state only after interpretation.

Figure 2 separates raw state-edge language from the narrower set of plausible specific state edges. Managed futures and trend-following products are the clearest current case: 18 of 90 documents contain plausible specific state-conditioning language. Crypto futures products also contain some specific state language, mainly around futures-market states such as contango and backwardation. Several other families have raw state-looking language, but little or no language that survives the

stricter audit.

Figure 2: Explicit state-conditioning is concentrated in a few product families



Notes: The figure uses the same 911-document systematic tier. A raw state edge is any extracted edge labelled as strategy-conditioned-on-state. A plausible specific state edge is a raw state edge that passes a deterministic audit for specific state language rather than generic market-condition language. The audit is used here as a measurement diagnostic, not as a final human-reviewed label.

This descriptive pattern motivates a broader product–research comparison than state-conditioning alone. For many product families, the natural connection to academic finance is a strategy, instrument, payoff rule, benchmark, risk, or mechanism. The product–research link therefore asks what kind of academic object is closest to the product graph.

In the current 135-link review queue, 45 candidate links are classified as same-strategy links, 32 as mechanism links, 31 as broader-pattern links, 15 as concept-only matches, and 12 as rejected matches. These are first-pass review labels and should not yet be read as population shares.

[Placeholder] – replace this paragraph with final family-level evidence after source checks. The paragraph should distinguish strategy-level links, mechanism-level links,

market/instrument links, and boundary cases without treating the current queue as final evidence.]

Table 3: First-pass link labels differ by product family

Product family	Most common link	Strategy	Mechanism	Broader pattern	Weak / rejected	Reviewed links	Reading
Managed futures / trend following	Same strategy	12	0	0	0	12	Product and academic languages meet at trend following and time-series momentum.
Multifactor smart beta	Same strategy	12	0	0	0	12	Product families map naturally to factor investing and smart-beta academic objects.
ESG / sustainable screen	Same strategy	12	0	0	0	12	The link is mostly a strategy or screen, though individual papers may differ in mechanism.
Volatility exposure / control	Mechanism	0	12	0	0	12	Academic links run through VIX, variance-risk premium, and volatility-risk-premium mechanisms.
Inflation-sensitive allocation	Mechanism	0	11	1	0	12	The link is mainly inflation hedging and inflation-risk exposure, not a single named product strategy.
Real-asset inflation hedge	Mechanism	0	9	3	0	12	Real assets connect to inflation-hedging mechanisms and asset-class broader patterns.
Crypto futures	Broader pattern / market	0	0	12	0	12	Academic counterparts study bitcoin futures and futures-market structure more than the product strategy itself.
Crypto spot / reference	Broader pattern / market	0	0	12	0	12	The link is mainly shared asset/market exposure.
TIPS / real return	Broader pattern / market	0	0	3	0	3	Inflation-indexed bonds provide an instrument-level link; the current review set is small.
Quality / growth selection	Boundary	4	0	0	8	12	Retrieval often drifts to earnings quality or unrelated “quality” language.
Managed volatility overlay	Boundary	3	0	0	9	12	Some target-volatility links are useful, but generic risk-control retrieval is noisy.
Low-volatility / low-beta	Boundary	2	0	0	10	12	Academic retrieval often confuses low-volatility investing with flow volatility or minimum-variance mathematics.

Notes: Counts are from the current first-pass review queue. “Weak / rejected” combines concept-only matches and rejected matches. The table is provisional: labels are useful for organizing the link typology, but selected cases still need source-text verification before final claims.

Table 4 organizes representative product families by the level at which the product graph currently connects to nearby academic objects. Managed futures and multifactor smart beta are current examples of strategy-level links. Volatility and inflation products are current examples of mechanism or rationale links. Crypto futures are closer to market or instrument links. Quality and low-volatility are boundary cases where similar words can retrieve adjacent academic objects.

[Placeholder – after source-text audit, replace this placeholder with a concise statement of which representative examples survive as strategy, mechanism, market/instrument, or boundary links. The statement should use reviewed product excerpts and reviewed academic evidence, not only retrieval labels.]

[Placeholder – quality check before stronger claims: verify 2–3 source papers per

Table 4: Representative product–academic link types

Product family	Product graph signature	Nearest academic object	Link type	Main caveat
Managed futures / trend following	Futures-based rules; trend-following strategy; asset-class rotation	Time-series momentum; trend-following strategies	Same strategy	Need source-text check that the product rule is genuinely trend-following, not broad managed-futures branding.
Multifactor smart beta	Factor/style targeting; index-like rules; portfolio construction	Factor investing; smart beta; factor models	Same strategy	Academic work may study factor construction more generally than investable index implementation.
Volatility exposure / control	Volatility-linked instruments; exposure rules; VIX or variance-risk terms	Variance risk premium; volatility risk premium; VIX forecasting	Mechanism / rationale	Academic mechanisms may explain volatility-risk pricing rather than the product wrapper.
Inflation-sensitive allocation	Real assets; allocation rules; inflation-risk exposure	Inflation hedging; inflation risk; asset allocation under inflation	Mechanism / rationale	The link may be through inflation-hedging rationale, not a single named strategy.
Crypto futures	Futures contracts; bitcoin exposure; roll or term-structure language	Bitcoin futures; contango / backwardation; futures price discovery	Shared market / broader pattern	Academic work may study the market or instrument, not the product strategy.
Quality / growth selection	Stock-selection rules; quality or growth language; fundamentals	Quality factor; earnings quality; quality investing	Boundary / noisy	Retrieval often confuses investment quality with accounting earnings quality or generic quality language.
Low-volatility / low-beta	Factor/style targeting; defensive equity; index rules	Low beta; betting against beta; minimum variance	Boundary / noisy	Retrieval can drift to volatility of flows or mathematical minimum-variance problems.

Notes: The table selects representative product families to illustrate link types. Product graph signatures summarize the recurring product-side graph objects. Nearest academic objects summarize the FrontierGraph-side retrieval target. Link types and caveats are current review labels. A local source-evidence audit checks these links against product-side graph excerpts and FrontierGraph academic retrieval evidence; full paper-abstract verification remains a later quality-control step.

worked-example family and 2–3 product excerpts per worked-example family. The checks should confirm that the product graph, academic counterpart, and link label all survive source-text review.]

[**Placeholder** – review noisy families before circulation: quality / growth selection, low-volatility / low-beta, and managed-volatility overlay. These families are useful boundary cases, but they should not be described as final evidence without manual review.]

[**Placeholder** – classification decision: decide whether the current “broader pattern” label should be split into “same instrument or market” and “broad broader pattern.” Crypto and TIPS are the main cases where this distinction may matter.]

7.1 Four Worked Examples

Managed futures is the cleanest same-strategy example. The product graphs describe futures-based strategies, rules, and trend-following language. The academic retrieval

returns papers such as *Profitability of time series momentum*, *Time series momentum in the US stock market*, and *Optimal allocation of trend following strategies*. The link is close because the product and academic objects are both operational strategies: trend following and time-series momentum.

Multifactor smart beta gives a second same-strategy case. These product graphs are dominated by factor/style targeting, portfolio rules, and index-like implementation. The academic side retrieves factor investing and smart-beta papers. The product is sold as a rule-based factor implementation, and the academic neighborhood studies factor construction, return models, and factor-investing performance.

Volatility exposure is different. The best academic counterparts are not usually product-wrapper papers. They are mechanism papers around the VIX, variance risk premium, and volatility risk premium. This is a mechanism link: the product may use volatility-linked exposure or volatility-control rules, while the academic literature studies why volatility risk is priced and how volatility measures forecast returns or risk premia.

Quality and low-volatility show the boundary of the method. A keyword-only approach would treat many retrievals as academic support. The graph-and-review view is stricter. “Earnings quality” papers are often about accounting or bank reporting, not quality investing. “Low volatility” papers may study capital-flow volatility or fund-flow volatility, not low-volatility equity products. These cases are useful because they show that product–academic comparison needs typed links, not vocabulary overlap.

[**Placeholder** – rerun link review later with source-text or abstract evidence. The current link labels are first-pass labels based on titles, matched terms, retrieval tiers, product-family context, and deterministic risk flags.]

8 Interpretation

[**Placeholder** – write the final interpretation after the reviewed evidence is fixed. The interpretation should state the main empirical pattern in product–research links and separate source-checked findings from measurement lessons.]

The current evidence supports a more precise statement: product graphs are useful because they separate product design objects before any comparison with academic research is attempted. A product can mention volatility as a risk, target low-volatility

stocks as a style, follow a volatility-control rule, or use volatility-linked instruments. Those are different objects. A keyword screen tends to collapse them; a graph representation keeps them separate. The same logic applies on the academic side: a paper about earnings quality is not automatically a paper about quality investing, and a paper about capital-flow volatility is not automatically a paper about low-volatility equity products.

[**Placeholder** – add one paragraph explaining the professional interpretation: the method can help distinguish research-grounded strategy design from wrapper innovation or superficial theme language.]

[**Placeholder** – add one paragraph explaining the academic interpretation: the method can show which research objects travel into market practice and which product objects remain under-studied.]